

THE LEDGER FRAMEWORK

Valuation Manifesto

Governing Principles for Valuation and Profit-and-Loss — Subordinate to the Architectural Constitution and the Market Data Manifesto

Version 1.0 — July 2026

A valuation is a projection of the record, and each mark earns its place by proving itself from the one before.

Abstract

A valuation is not a bare number. It is the value of a unit read at coordinates — a ledger position, a unit state, a corporate-action frame, a market-data cut, and the models the price is bound to — and the same coordinates reproduce it to the last minor unit. This manifesto states the principles that make valuation and profit-and-loss a projection of the one record and nothing beside it. Two layers, kept apart on the first page, carry the whole design: each model-priced leg *re-enters as a recorded observation*, bound to its model by lineage (MD-15); the net asset value, the profit-and-loss, and the valuation chain that assembles them are a *projection* of the record, storing nothing (C-8.2). A unit's current value is a *valuation chain*: each mark is proven from the last by a profit-and-loss explain whose residual sits below a declared bound, and an unexplained move is a broken chain, forbidden. Every mark is dispute-ready, settled by replay, extending the Market Data Manifesto's exhibit list. A corporate action is a valuation sandwich, its frame re-coordination a zero-profit identity and its near-zero residual the proof it went well. A risk report is the same valuation on shifted market data — no second system, no second code path. Each principle is an *article* derived from the seven governing principles the two parents already fixed — auditability, reproducibility, time travel, correctness, order, simulability, and the derived dispute-readiness — and rests on the clause of a parent that carries it. The document is rated against both parents, never the other way round; where it conflicts with either, the parent prevails.

1. Valuation in the Ledger's Terms

A **valuation** is the value of one unit, together with its sensitivities, read at a fixed set of coordinates. It is never the truth about what the unit is worth; it is what a stated model says the unit is worth given stated inputs, exactly as the Market Data Manifesto holds an observation to be a fact about what a source said, not about what a thing is (MD-1). The value of the whole book is a sum of such per-unit valuations over the owned coordinate (C-4.10, C-8.2).

Two layers, and they must never be confused. A stored per-unit model price, or a stored greek, is a **re-entered observation**: the ledger runs no model (C-14.9), so a model is evaluated outside the fold and its output re-enters through the one door as a recorded observation (C-14.15), bound to that model by lineage (MD-15), and stale the moment an input it consumed moves (MD-8). The net asset value, the profit-and-loss, and the **valuation chain** that assembles them are of the other kind: a **projection** that computes over the record — owned quantity

times price, summed over valued units — storing nothing and rebuilding from the record alone (C-8.2, C-14.11). The projection consumes the model-priced legs as *leaves* (MD-12), exactly as it consumes a directly observed stock quote: a plain observation enters the same sum. So “valuation is a pure function of ledger state” is a true statement about the projection *recipe*; “a valuation re-enters as an observation” is a true statement about each model-priced *leg*. Both hold at once, at their own layers, and every principle below follows from the pair. The distinction is not pedantry; it is where a defect is repaired. A wrong leg is a mismarked observation, re-derived with its model; a wrong total is a mis-assembled projection, rebuilt from the record. Confuse the two and you repair the wrong thing.

Vocabulary, mapped once. This manifesto uses the fixed names of both parents and coins no synonym for any of them (C-Auth.4). It reuses *unit*, *wallet*, *balance*, *projection*, *the immutable log*, *the market data operator*, *profit-and-loss explain*, *explain item*, and NAV; *observation*, *re-entered observation*, *frame*, *cut*, *as-of/as-at*, *complete lineage*, and *simulated path* come from the Market Data Manifesto unchanged. It coins four names, each checked against the fixed set: the **valuation record**, the **valuation chain** (always two words — the immutable log alone is hash-chained, C-4.8, and “chain” bare is reserved for it), the **profit-and-loss explain certificate**, and the **corporate-action valuation sandwich**. A scenario perturbation is a **shift** — never a “shift operator”, since *the market data operator* names one thing only, the corporate-action transform of C-9.2 (MD-13). A scenario is a *simulated path* under a recorded shift (MD-11); no “derived world” is coined.

2. The Articles

Each article derives from one or more of the seven governing principles and rests on a clause of a parent — the Constitution (C-*sec.n*) or the Market Data Manifesto (MD-*n*). The identifiers VM-*n* are append-only: never reused, never renumbered.

VM-1. A valuation is a re-entered observation of a unit’s price; the book value is a projection over such observations. VM-1 The two layers of Section 1 are the first article, because every other rests on them. Where a price is read directly from the record — a listed close — it is a plain observation. Where a price is produced by a model, the ledger does not run the model (C-14.9); the model runs outside the fold and its output, together with the sensitivities computed with it, re-enters through the one door as a recorded observation bound to that model (C-14.15, MD-15) — the **valuation record**. The net asset value is then a projection: replay the log to recover the wallet’s composition, multiply by the current price vector over the valued units, and store nothing (C-8.2, C-8.3). What the projection multiplies is a mixture of plain and re-entered observations; it treats them alike, as leaves (MD-12). No valuation adds a store to the ledger: it is a further fold of the same log (C-1.4).

(*Correctness, Reproducibility; C-8.2, C-14.9, C-14.15, MD-6, MD-15.*)

VM-2. A valuation is meaningless without its coordinates; the same coordinates reproduce it bit for bit. VM-2 A number alone is not a valuation. A valuation is fixed only by its coordinates: the ledger position it values — the composition at an *as-of* world-moment, recorded at an *as-at* cut (C-2.7, MD-4); the lifecycle state of the unit, which pricing depends on and which is read from the ledger, never from a store of its own (C-8.3); the corporate-action **frame** the value lives in (MD-13); the market-data **cut** that fixes which observations are in force (MD-6, MD-12); and the models the price is bound to (MD-15). The position’s as-at cut and the market-data cut are two coordinates, not one, and independently selectable: hold the position at yesterday’s as-at and move the market-data cut to today, and you have yesterday’s book on today’s prices — a distinct, legal query on which profit-and-loss attribution and the shifted worlds of VM-10 both rely. Given the same coordinates, any party recomputes the identical

valuation, to the last minor unit (C-2.2). Reading a recorded valuation back is unconditional; *re-deriving* it from its inputs needs the retained model — the one coordinate whose retention is external to the record, a governance matter (C-14.15, §4). Because each input is pinned at its own cut and never to a mutable store, a valuation struck “mid-update” cannot arise (MD-8, MD-12): the state a valuation reads is the folded state, whole, or the valuation is not struck. A valuation carried without its coordinates is not a weaker valuation — it is not one.

(*Reproducibility, Time travel, Correctness; C-2.2, C-8.3, MD-4, MD-13.*)

VM-3. A unit’s value is a valuation chain; each mark is proven from the last. The VM-3 current valuation of a unit is not an isolated number but the head of a **valuation chain** — an ordered sequence of valuation records, each proven from the one before it by a profit-and-loss explain (VM-4). The chain is append-only: a re-mark is a new link — itself a re-entered observation (VM-1) — forward, never an edit of an old one (C-4.8, C-12.4), and the value the book showed at any past cut survives unrewritten beside the current one (C-12.1). The greeks that close one link seed the next: today’s exit sensitivities are tomorrow’s entry sensitivities, so the explain carries measured sensitivities at both ends of every link. Measured endpoint greeks are not a convenience: they keep the first two orders exact and squeeze any gap in Hessian coverage — the cross-greeks an implementation did not populate — into the third-order term and the residual, rather than smearing it through delta and vega, where it would corrupt the very hedge the greeks are supposed to size. The chain is the valuation layer’s instance of complete lineage (MD-6); it reaches back through the price vector and the operator-adjusted inputs to the leaf observations. A unit may carry more than one valuation chain at once — one per re-marking convention, cadence, or desk — and they do not compete: each decomposes the one change in net asset value (C-8.4) and localises to its own recorded convention coordinate (VM-5).

(*Order, Auditability; C-4.8, C-12.1, C-12.4, MD-6.*)

VM-4. The profit-and-loss explain is the certificate, and it states honestly what is VM-4 **exact.** Each link carries a **profit-and-loss explain certificate**: a decomposition of the value change between two marks, read the way a desk reads it. Carry first — the value the position sheds or earns as a day passes at unchanged market. Then the delta line — the move read against the entry delta. Then the gamma-and-theta balance, the core of the daily explain: convexity earns on the move, decay pays for it, and a delta-hedged position turns a profit only once the move clears its breakeven, the point where the day’s gamma gain just covers the day’s theta. Then vega, bucket by bucket — by name, by expiry, by strike, never a single number — with the cross-greeks vanna and volga among the higher-order terms. Then the lifecycle and state changes the link crossed. And last, one explicit residual (VM-6). Two forms of the market lines exist and the certificate must not conflate them. The *ex-ante* estimate — the reading available *before* the move, computed from the entry greeks alone — is $\Delta \cdot \delta + \frac{1}{2} \delta^T \Gamma \delta$, the delta-and-gamma a desk pre-computes, and carries a *deterministic, signed* gap to the reprice equal to the variation of gamma over the move. The *exact* decomposition uses the greeks at *both* ends: the value change is the move times the average of the entry and exit deltas, less a one-twelfth correction for the gamma that moved (a both-endpoint quadrature correction), plus a residual R that is identically zero only for a book cubic in its risk factors. Using entry greeks in place of both-end greeks “is not a smaller model; it is an inconsistent one”, so the certificate names which form it used. The residual behaves well only under conditions: for a value smooth to fourth order, $R = O(\|\delta\|^4)$ — doubling the move multiplies its leading part by sixteen; at a digital, a barrier, an autocallable, any payoff that is not smooth, $R = O(1)$, the size of the discontinuity, the instant a move straddles the feature; and the calendar direction is never covered by that bound — a day of carry near expiry lands in R at full size. The certificate therefore *never* promises a universally bounded residual. The total the certificate explains is path-independent: it is exactly the change in net asset value (C-8.4). Only the *decomposition* depends on the marks chosen along

the way and on the convention (VM-5) — the identity that the parts sum to ΔNAV holds on every path; the split into parts does not, and no second running total of realised profit-and-loss is kept beside NAV to disagree with it (C-8.4). Because the residual is the balancing term — the value change less every attributed line — the certificate reconciles to the change in net asset value by construction, not by coincidence: it is a projection of that one number, and no split it prints can make it disagree with the fold. Attribution decomposes ΔNAV ; it never recomputes it. This explain *composes* with the Constitution’s reordering explain: there is one profit-and-loss explain, and a reordering-refold line (C-12.6) and a market-factor line here are two disjoint kinds under one roof, each attributed to exactly one cause. The certificate extends the taxonomy; it does not re-coin the term.

(*Correctness, Auditability, Order; C-8.4, C-12.6, MD-5.*)

VM-5. Attribution is convention-dependent; the convention is a declared, recorded term. VM-5 How a move splits across the greeks is not unique. Where a risk factor is not a market map of its driver — an implied basket correlation is the canonical case — the split between the spot channel and the vol channel depends on which quantity the desk holds invariant as the others move, and two conventions that reproduce today’s marks exactly differ in tomorrow’s attribution: same marks today, different *attributed* profit-and-loss tomorrow. That invariant is a declared, recorded term of the valuation (MD-6). Two consequences follow. The same convention reproduces the same attribution, bit for bit, a deterministic function of the record. And a dispute about attribution is a dispute about convention: replay *localises* it to the differing term (MD-14) but does not adjudicate which convention is right, which is model choice and out of scope (MD-9, MD-15). The manifesto therefore never speaks of *the* attribution as unique, nor of a shadow greek — the value routed through the held-invariant factor — as a property of the market: under the engine default that holds the at-the-money volatility constant, every spot-channel shadow number is identically zero, and must be flagged as computed under that convention, never quoted as a market fact.

(*Correctness, Reproducibility; MD-6, MD-9, MD-14, MD-15.*)

VM-6. “Explained” means the residual sits below a declared bound; a breach is a recorded fact, never absorbed. VM-6 A move is *explained* when its residual R falls inside a declared tolerance. The tolerance is a declared term, not a universal constant: a valuation is certified only when each instrument’s residual is within its bound and the aggregate is within the aggregate bound, and those bounds are recorded in the model configuration, mapping between price space and vol space through vega. These bounds roll up: the per-unit residuals aggregate to a book-level residual against the aggregate bound, where cross-unit and correlation profit-and-loss — routed under the declared convention (VM-5) — must also come to rest. The residual is itself a re-entered observation — a recorded diagnostic, stale if its inputs later move, never a silent pass (MD-9, MD-15). It is a sensor, not a rounding error: a residual that is small, mean-zero, and regime-independent certifies the bookkeeping is a control variate — a correction whose smallness can be trusted —; a residual that is structurally large, sign-biased, or wakes up in stress reports that the honest attribution of this position is a scenario, not a greek, and demands re-mark or model review. Two things the bound must survive. First, the anti-plug: that the certificate reconciles to the change in net asset value proves *nothing* — the residual is the balancing term (VM-4), so the parts sum to ΔNAV whether the move is explained or not; only the residual *below its declared bound* certifies. Reconciliation is automatic; explanation is earned. Second, the bound’s own adequacy is a governance assumption, named here as the parent named the identifier grain and the delivery frame: a bound set too loose makes every move “explained,” fires no breach, and empties “unexplained profit-and-loss is forbidden” (VM-7) in silence. The guard is that the residual is recorded whatever the bound — its sign and regime behaviour, not its level alone, are inspectable after the fact — and that the bound’s calibration is itself a

recorded, reviewable event, so a loosened bound is an auditable change, never a silent one. A breached bound is booked — a named line in the profit-and-loss explain (MD-5) — and acted on. A bookkeeping that announces when to stop believing it is worth more than one that is elegant while it lasts.

(*Correctness, Auditability; C-2.4, C-13.2, MD-9, MD-15.*)

VM-7. Unexplained profit-and-loss is a broken chain, and a broken chain is forbidden. VM-7

A link whose residual exceeds its declared bound, or whose leaf inputs have moved under it, is not a link: it is a broken valuation state, and a broken state cannot hide (MD-8). Two mechanisms keep it visible. A projection over a corrected leaf recomputes on the next read, so a stale *projection* cannot arise. A re-entered valuation record that consumed a corrected or superseded input — a wrong price, a restated corporate action behind its frame — cannot re-run itself (C-14.9); its lineage marks it stale and it stands as a flagged open item for re-derivation (MD-8, MD-10), read through a projection that carries the staleness flag, never as the raw stored number. Staleness propagates forward along the chain: a superseded link flags stale every downstream link whose certificate chained from it, and each re-proves *forward from the superseding predecessor* — never editing the old proof — so a certificate can never keep attesting a transition from a mark that no longer heads its segment (MD-6, MD-8). The as-known branch survives beside the corrected one (C-12.1), so the chain resolves into the two honest answers rather than silently forking. Nothing in the chain is repaired by overwriting; a correction repairs forward, and where money already moved it moves back only under authorised compensation (C-12.4). What is forbidden is the *silent* unexplained move: an unexplained profit-and-loss is a defect the record must surface, not a number the book may carry.

(*Correctness; C-1.5, C-14.9, MD-8, MD-10.*)

VM-8. Every valuation is dispute-ready. The current valuation of every unit is dispute- VM-8 ready: its chain, the certificates along it, and its coordinates suffice to reproduce and defend it, bit for bit, against any counterparty — a valuation dispute and a collateral dispute alike. This is the Market Data Manifesto’s dispute-readiness (MD-14) reached at the valuation layer, and it *extends* that manifesto’s exhibit list: to the observation, its provenance, the cut, the frame, and the bound model, valuation adds the **valuation chain** and its **certificates**, exhibited as further rows of the one replay, never a second replay doctrine. Its bound is exact and inherited: replay settles whether the mark is the faithful, deterministic consequence of the inputs the record held at its stated coordinates, the as-at pinned to the mark’s own cut so the number that stood then is reproduced, not a later one. It does not settle a two-sided economic dispute — a counterparty’s mark, from its own surface and convention, replays bit for bit too — but exhibiting the chain and the convention *localises* the disagreement to the differing input, frame, model, or convention, which is the value replay delivers (VM-5, MD-14). A collateral dispute is bounded the same way: replay settles the *mark* inside the call, never the collateral-agreement terms — haircut, eligibility, netting — which are model choice and out of scope (C-Scope.11).

(*Dispute-readiness (Auditability + Reproducibility); C-2.1, C-2.2, MD-14, MD-15.*)

VM-9. A corporate action is a valuation sandwich; its frame re-coordination is not profit-and-loss. VM-9 Every corporate action that changes a unit’s state triggers a **valuation sandwich**: a valuation struck immediately before the transition and another immediately after, with a full profit-and-loss explain across the pair. Across the sandwich the state and the coordinate frame change discontinuously, so the smooth explain of VM-4 does not apply — there is no smooth move to expand along. The only admissible explanation of the jump is the market data operator (C-9.2, MD-13): it transports the pre-frame mark into the post-frame, acting on leaf observations, with derived objects recomputed from operator-adjusted inputs, never scalar-transported. The frame re-coordination itself is a *zero-profit identity*: the same value in

new coordinates, the consistency-of-reference that forbids multiplying a new share count by a stale price (C-11.3). Only genuine economics leaving or entering the position — a special dividend paid away — is profit-and-loss, and it earns its own explain line; there is no “corporate action profit-and-loss” beyond it. That the sandwich residual is ≈ 0 is the proof the action went well, and $\approx$ means exactly three declared tolerances and no more: the minor-unit rounding applied once at read (C-4.6), the declared convention’s own rounding, and the calibration tolerance of any re-derived surface or curve (VM-6). A residual materially above these is not noise — it says the operator or the delivery frame was mis-declared, which replay localises (MD-14) and MD-8 and MD-10 handle as a superseding event demanding re-derivation. The operator exists only once the action’s terms are *resolved*, not merely announced; before resolution the frame is provisional and the sandwich is struck as such (MD-13). A third timing case is a resolved action that reaches the door *late*, after valuations were already chained across its ex-date. This is a reordering, not a lineage supersession: the intervening links never carried the action in their lineage, so they are flagged stale by the reordering trigger (C-2.7, MD-5), not by the frame-supersession path of VM-7. The action takes its place at the ex-date, its sandwich is struck *retroactively* there, and the spurious market-move line the chain had booked — a two-for-one split reaching the door three days late reads as a fifty-percent crash until it arrives — is reclassified as the zero-profit frame re-coordination, the reordering carried as its own line in the profit-and-loss explain (C-12.6). The loss is never left standing.

(*Order, Correctness, Time travel; C-9.2, C-9.3, C-11.3, MD-13, MD-5.*)

VM-10. Risk is valuation in simulated worlds. A risk report is not a separate system and not a second code path: it is the same per-unit valuation recipe of VM-1 applied to *shifted* market data. The Constitution already fixes the doctrine — “we live in a simulation: production is simply the one path whose events happened to be real” (C-2.8) — and the Market Data Manifesto that simulated market data is real market data under a different seed (MD-11). A scenario is a simulated path in which one or more observations are moved by a recorded **shift**; a greek is a sensitivity read across the base and shifted paths. Shifts *compose* — spot up twenty percent, volatility down, in any combination — each composition yielding a new vector of valuations and greeks by re-application of the one recipe. Base and simulated worlds share their units, their models, and the market data operator; only the market-data state differs. And the gap between the greek approximation and a full revaluation is itself a computable, recordable fact — the ex-ante estimation gap of VM-4: the deterministic, signed gamma-variation term *plus* the residual R (VM-6), small for a smooth book and $O(1)$ where the payoff is not — so a sound system computes both and records their difference, on the same terms as a production day, never choosing silently between them. The shift is recorded, exactly as a seed is (MD-11): it is the single non-record input of the scenario, so the path replays. But a path that recorded only its shift could not replay — every model-priced valuation along it re-enters as an observation like any other (VM-1), into the simulated path’s *own* record, never as a link in the real unit’s valuation chain — so a stress run is a derived object with full lineage, dispute-ready on the same terms as production (VM-8). And because risk and the books run the one recipe, the risk report and the official profit-and-loss cannot disagree by construction: the classic risk-versus-books reconciliation break — two systems, two code paths, two numbers — has no way to arise.

(*Simulability, Reproducibility; C-2.8, MD-11, C-14.11.*)

3. The Life of a Valuation

One unit walks the whole picture. Follow a single long call, struck at 100 on a stock at 100, three months to expiry, marked at a 30% implied volatility. Its value is 5.98, its delta 0.53, its gamma 0.0265, its decay -0.047 per trading day. Each row of Table 1 is a link; each is proven from the

one above.

Link	What changed	Explain into this mark
Base	struck at all coordinates	value 5.98; delta 0.53, gamma 0.0265
Carry day	one trading day, spot flat	carry -0.047 ; residual $< 0.001 \Rightarrow 5.93$
Market move	spot 100 $\rightarrow$ 102, vol held	delta $+1.060$, gamma $+0.053$; ex-ante residual -0.001 (bound 0.010) $\Rightarrow 7.09$
CA sandwich	2:1 split; strike $\rightarrow$ 50, deliverable doubles	frame re-coordination (zero-profit); economics 0; residual $\approx 0 \Rightarrow 7.09$

Table 1: A valuation chain of four links on one unit. Each link isolates one effect for legibility; on a real book carry and market move fall in a single link and both appear as explain lines. The market-move residual is the ex-ante gap (the higher-order and gamma-variation content), well inside its declared bound. The corporate-action link is a sandwich: the pre-ex value 7.09 in the old frame equals the post-ex value 7.09 in the new one, and the near-zero residual is the proof the split was applied consistently — not a phantom valuation from a new share count against a stale price (C-11.3).

The market-move link also reads as the gamma-and-theta balance of VM-4: the \$2 move clears this option’s breakeven of \$1.89, so a delta-hedged holder’s gamma gain (0.053) just covers a day’s decay (0.047). Inside the breakeven the day loses; outside it the day earns; and that balance, not the delta, is what a long option position is really holding.

A shifted-world column. From the base mark, a risk report is one more valuation under a recorded shift. Shift spot $+20\%$, to 120, and re-apply the same recipe. The greek estimate is $0.53 \times 20 + \frac{1}{2} \times 0.0265 \times 20^2 \approx +15.9$; the full revaluation is $+14.9$ (the mark moves to 20.9). The gap, $\approx +1.0$ — near a fifteenth of the move, because a twenty-percent shift is far outside the regime where the quadratic is a good control — is not an error to be suppressed but a recorded diagnostic (VM-10). The shift composes: add a volatility-down leg and the same recipe returns a new vector, no second code path.

A link that breaks. Now swap the call for a digital that pays a fixed coupon if the stock is above 100 at expiry, and take a link that straddles the strike a week out. The two-point explain fails on sight: the reprice is the whole coupon, while the endpoint deltas, evaluated where the value has already flattened either side of the strike, read near zero, so the entire notional lands in the residual — $O(1)$, not $O(\|\delta\|^4)$. The certificate does not hide this: the residual breaches its bound, the breach is booked as a named line (VM-6), and the chain is flagged broken until the position is re-marked (VM-7). The honest attribution here is a gap scenario with a probability attached, not a greek — which is exactly what VM-4 warned the smooth explain could not see, made visible by the very residual it refuses to absorb.

Disputed. A counterparty contests the collateral mark. It is not argued: replay exhibits the chain, its certificates, and its coordinates, and reproduces the mark bit for bit, the as-at pinned to the mark’s own cut (VM-8, MD-14). Replay cannot decide whose surface or convention is right, but it localises the disagreement to a differing input, frame, model, or convention.

4. What This Manifesto Does Not Govern

This manifesto governs how valuation and profit-and-loss are recorded, proven, and projected. It does not govern **which model, surface, prior, or re-marking convention is correct** — model choice, whose test is hedging performance, out of scope (C-Scope.11, MD-9, MD-15); nor **model governance, validation, and retention** — a valuation reproduces from recorded prices unconditionally and from a model’s number only if the model is kept (C-14.15); nor the

numerical methods by which a model is solved — regression, quadrature, surrogate pricing — which are the reference implementation’s, not the specification’s (C-14.9); nor **price formation**, the **terms of a corporate action** (they arrive as recorded events, MD-13), or **settlement**. These lie outside the Ledger boundary and are reconciled against it only at its perimeter.

5. Subordination

This manifesto is subordinate to two parents — the Architectural Constitution and the Market Data Manifesto — and rated against both, never the other way round. Its vocabulary is theirs: one name per component, and no synonym for any fixed term (C-Auth.4). Where a principle here would conflict with either parent, the parent prevails and the conflict is parked with the exact amendment text and the exact clause it replaces — never resolved by quiet narrowing, which for a guarantee of either parent is the worse failure. A genuine departure amends this document explicitly first; an article identifier VM- n is never reused or renumbered.

Conclusion

Valuation has the same root cause of failure as everything else the Ledger touches: the same fact held in two places, marked by two logics, drifting apart. The Ledger removes it here as everywhere. A valuation is a projection of the one record; each model price it consumes is an observation; each mark it produces is proven from the last by a profit-and-loss explain whose residual it must keep below a declared bound, and an unexplained move is a broken chain, not a number to carry. Every mark carries its coordinates, is dispute-ready by replay, survives a corporate action as a frame re-coordination that is not profit-and-loss, and is re-run in a simulated world for risk by the very same recipe. Operate it any other way and the divergence the architecture exists to remove comes back; this way, a valuation is a fact, and there is nothing left to reconcile.